(a) Bottom-up

Model

Proof

Model–proof mismatch

Main theorem

```
theorem smd_convergence
  ... := by
```

Test interfaces

Theory rework

Stochastic interfaces

```
def filtration ...
lemma noise_centered ...
```

Lean foundations

```
def MeasurableSet ...   theorem integral_add
```

Measure theory (Ω, Σ, μ)

Integration $\int f \, \mathrm{d}\mu$

Probability $\mathbb{E}[X]$

Stochastic processes $X_0, X_1, \ldots$

Convex analysis $\nabla f(x)$

(b) Proof-obligation-driven (ours)

Main theorem & proof

```
theorem smd_convergence
  ... := by
have h_noise : Proof obligation
 condExp F (ε x) = 0 := ?_
```

Theory under construction

Next theorem

```
import
SOptLib
```

Reuse

prove

Sampling history

```
def filtration
  seq t := ...
```

Stochastic oracle

```
def oracle
```

$\varepsilon(x) = G(x, \xi) - g(x)$

$G(x, \xi)$

x

Bridge lemmas

```
lemma
noise_centered :
 condExp F (ε x)
 = 0 := by ...
```

Extract, Generalize & Integrate

SOptLib

Definitions

Interfaces

Lemmas